PRESIDENCY UNIVERSITY

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CSE3146 – Advanced JAVA Programming

LAB SHEET – 3 Activity 1

Module 3 – Java Database Connectivity

Q1. Write the useful classes , interface and methods available in sql.*

Solution : Principal JDBC interfaces and classes under sql.*

Let's take an overview look at the JDBC's main interfaces and classes which we'll use in this article. They are all available under the *java.sql* package:

- **Class.forName() :** Here we load the driver's class file into memory at the runtime. No need of using new or creation of object.
- **DriverManager:** This class is used to register driver for a specific database type (e.g. Oracle Database in this tutorial) and to establish a database connection with the server via its **getConnection()** method.
- **Connection:** This interface represents an established database connection (session) from which we can create statements to execute queries and retrieve results, get metadata about the database, close connection, etc.

- **Statement** and **PreparedStatement**: These interfaces are used to execute static SQL query and parameterized SQL query, respectively. **Statement** is the super interface of the **PreparedStatement** interface. Their commonly used methods are:
  1. **boolean execute(String sql):** executes a general SQL statement. It returns *true* if the query returns a *ResultSet*, false if the query returns an update count or returns nothing. This method can be used with a *Statement* only.
  2. **int executeUpdate(String sql):** executes an INSERT, UPDATE or DELETE statement and returns an update account indicating number of rows affected (e.g. 1 row inserted, or 2 rows updated, or 0 rows affected).
- **ResultSet executeQuery(String sql):** executes a SELECT statement and returns a *ResultSet* object which contains results returned by the query.
- **ResultSet:** contains table data returned by a SELECT query. Use this object to iterate over rows in the result set using next() method.
- **SQLException:** this checked exception is declared to be thrown by all the above methods, so we have to catch this exception explicitly when calling the above classes' methods.

Q2. Use MySql to Create a database userdb and a table user with attributes

id, password, Fullname, email

All are string.

Make id as primary key

Solution :

Step 1: Use mysql to create the database userDB and table user

Create a database userDB

CREATE DATABASE USERDB;

Create a table user

```
CREATE TABLE USER(
id varchar2(30) NOT NULL PRIMARY KEY,
pwd varchar2(30) NOT NULL,
fullname varchar2(50),
email varchar2(50)
);
```

Q3. Using JDBC perform the following database operations for user table created above

i) insert
ii) update
iii) delete
iv) select

Algorithm :

Step1: GET the DRIVER ready

Import *java.sql*.* to use JDBC interfaces and classes
Invoke Class.forName("driver name);
Mysql driver
com.mysql.jdbc.Driver for mysql version below 8
com.mysql.cj.jdbc.Driver for mysql version 8 and above

Step2: GET the Connection path

Invoke DriverManager.getConnection("url to database", "username","password")

For MySql

```
Connection con = DriverManager.getConnection("jdbc:mysql://localhost:3306/db", "username","pwd");
```

Step3: GET the SQL prompt through createStatement() or prepareStatement()

```
Statement sprompt = con.createStatement(); // for static query
PreparedStatement dprompt = con.prepareStatement("dml query"); // for dynamic query
```

Step4: Execute Query through SQL Prompt

```
int  result=sprompt.execute("DDL query");  or
ResultSet rs=sprompt.executeQuery("DQL query");  or
int result = sprompt.executeUpdate("DML query");  or
int result = dprompt.executeUpdate();  // for DML
```

Step5: Process ResultSet or result to console

Solution :

i. Implementing Insert Statement

```
// Java program to illustrate inserting to the Database
import java.sql.*;
public class TestInsert
{
 public static void main(String args[]) throws Exception
{
  String id = "20211cse001";
  String pwd = "Forest#321";
  String fullname = " Rama Krishna K";
  String email = "ramakrishna@pu.in";
  Class.forName("com.mysql.jdbc.Driver");
  Connection con = DriverManager.getConnection("jdbc:mysql://localhost:3306/userinfo", "root", "afroz");
  Statement stmt = con.createStatement();
// Inserting data in database
  String q1 = "insert into userid values('" +id+ "', '" +pwd+
                 "', '" +fullname+ "', '" +email+ "')";
  int x = stmt.executeUpdate(q1);
  if (x > 0)
   System.out.println("Successfully Inserted");
  else
   System.out.println("Insert Failed");
         con.close();

  }}
```

Output :
Successfully Registered

ii) Implementing Update Statement

```
// Java program to illustrate
// updating the Database
 import java.sql.*;
public class Testupdate
{
public static void main(String args[]) throws Exception
{
String id = "20211cse001";
String pwd = "Forest#321";
String newPwd = "Afroz123";
 Class.forName("com.mysql.jdbc.Driver");
Connection con = DriverManager.getConnection("jdbc:mysql://localhost:3306/userinfo", "root", "afroz");
Statement stmt = con.createSttement();
// Updating database
String q1 = "UPDATE userid set pwd = '" + newPwd +
      "' WHERE id = '" +id+ "' AND pwd = '" + pwd + "'";
int x = stmt.executeUpdate(q1);
if (x > 0)
   System.out.println("Password Successfully Updated");
else
   System.out.println("ERROR OCCURED :(");
con.close();

 }}
```

Output :
Password Successfully Updated

iii) Implementing Delete Statement

```
// Java program to illustrate deleting from Database
import java.sql.*;
public class Testdelete
{
 public static void main(String args[]) throws Exception
{
  String id = "20211cse003";
  String pwd = "Forest#321";
  Class.forName("com.mysql.jdbc.Driver");
  Connection con = DriverManager.getConnection("jdbc:mysql://localhost:3306/userinfo", "root", "afroz");
  Statement stmt = con.createStatement();
  // Deleting from database
  String q1 = "DELETE from userid WHERE id = '" + id +
       "' AND pwd = '" + pwd + "'";

   int x = stmt.executeUpdate(q1);
   if (x > 0)
    System.out.println("One User Successfully Deleted");
   else
    System.out.println("ERROR OCCURED :(");
  con.close();
          }
  }
```

Output :

One User Successfully Deleted

iv) Implementing Select Statement

```
import java.sql.*;
public class Testselect
{
 public static void main(String args[]) throws Exception
 {
  String id = "20211cse001";
  String pwd = "Afroz123";
  Class.forName("com.mysql.jdbc.Driver");
  Connection con = DriverManager.getConnection("jdbc:mysql://localhost:3306/userinfo", "root", "afroz");
  Statement stmt = con.createStatement();
 // SELECT query
  String q1 = "select * from userid WHERE id = '" + id +
                    "' AND pwd = '" + pwd + "'";
  ResultSet rs = stmt.executeQuery(q1);
  if (rs.next())
  {
    System.out.println("User-Id : " + rs.getString(1));
    System.out.println("Full Name :" + rs.getString(3));
    System.out.println("E-mail :" + rs.getString(4));
```

```
 }
 else
 {
   System.out.println("No such user id is already registered");
}
 con.close();
  }
}
```

Output :

User-Id : 20211cse001
Full Name : Rama Krishna K
E-mail :ramakrishna@pu.in

Record Writing

Note: Submit the record on or before the due date.

Both Soft and Hard copies are required to be submitted.

RECORD WRITING INSTRUCTIONS

1. Solve the programming exercise using any IDE (Laptop / Mobile) or using any online compiler.
   A. Students can use online compiler or any preferable platform for the execution. Suggested is to use JDoodle. https://www.jdoodle.com/online-javacompiler/ Do test this site before your CA.
   B. Mobile users, kindly install JStudio - ide for java https://play.google.com/store/apps/details?id=com.qamar.ide.java&hl=en. This instruction is already given for solving your lab programs. If you haven't done, please do install, and test the app as soon as possible.
2. While solving your programming exercise, write the code in A4 sheet paper/Record. While writing on the paper, please add these info. "Presidency University" "Department of CSE" "Odd semester 2021-2022" "MODULE - 3" "Course code : CSE 3146", Course name : Advanced Java Programming, ID: ,NAME:_____, SEC:____ , Date:_____
   A. While coding (the soft copy) & writing in the paper/record, all your **CLASS NAME** and the **METHOD NAME** must be appended with your **LAST FOUR DIGIT student ID**. This is mandatory, even while WRITING in the paper/record.

For example: If your Registration number is 2021CSE0161 then

```
class sample0161   {
        void Method0161(parameterlist)  {
                //method body
        }
```

B. Take a screenshot of **your program & the output** from your mobile/laptop.

C. Take a photo of the handwritten program.

D. Put together (**4. A,B,C**) , **combine as one pdf**, with the file name as your student **registration number(ex. 2021CSE0161.pdf)**, and upload the file in Camu.

E. The document must be uploaded within the specified time in Camu.

Kindly follow the instructions very carefully so that your submission will be valid.